(a) End-to-end evidence synthesis

Question & query / retrieval → **Screening** → retained, screened full texts → **Coding** → evidence-linked estimates → **Synthesis** → **Verified artifacts + review status**

(b) Screening

Record input: JSONL / CSV / bibliography

Title • abstract • metadata
Dates • keywords • native MeSH

Title / abstract screening
Five dimensions / review criteria

S / P or needs full text

Main PDF + SI: paper_pool
Parsed hit: reuse
Miss: acquire + MinerU

Full-text screening
Evidence / coverage checks

Retained + screened
S / P per active policy

Unresolved
Review / resume

Missing source ≠ exclusion.

(c) Coding

Document input: primary full text + codebook

Verified source text / blocks
Question + effective codebook

A Candidate extraction
LangExtract / explicit LLM

candidates + exact source evidence

B Semantic finalization
Source-bound fields + decisions

Keep • correct • reject • pending

Estimate	Context	Evidence
point + stat.	cohort / group	quote + locator

Evidence-linked table + original values / review flags

Full A+B default; explicit ablations only.

FT + Coding share verified source caches.

(d) Pi Synthesis

Table + methods + primary evidence

Profile • SQL • evidence
Inspect table / staged sources

Commit common estimand + plan
Select • transform • dependence

Write code
Python / R

bounded feedback

Sandbox execute
+ validate

Finalize verified execution
Pooled estimate + uncertainty
Code • ledger • plots • report

Pending explicit; execution ≠ scientific validation.

Whole SR methods may expose results.

Shared context: research question • review protocol • codebook • analysis settings